

ISRAT JAHAN TULIN

[linkedin.com/in/israt-jahan-tulin](https://www.linkedin.com/in/israt-jahan-tulin) · github.com/Tulin206 · israt_jahan.tulin@outlook.com
<https://tulin206.github.io/> · +49 1782141007 · Dresden, Germany

MACHINE LEARNING ENGINEER | APPLIED AI & MLOPS

Machine Learning Engineer and Applied AI researcher with experience developing reproducible ML workflows for complex scientific imaging, sensor, time-series, and multimodal datasets. Skilled in PyTorch, TensorFlow/Keras, Scikit-learn, data preprocessing, feature development, model validation, and MLOps practices using Git/GitLab CI/CD, DVC, Docker/containers, unit testing, and HPC/GPU environments. Experienced in interdisciplinary projects involving tomography reconstruction, anomaly detection, object detection, autonomous transport perception, and sensor-based AI. While my deployment experience has been mainly in HPC and containerized research environments rather than public cloud, I have a strong foundation in scalable Linux-based ML workflows and am actively prepared to transfer these skills to AWS/Azure/GCP-based ML deployment.

TECHNICAL SKILLS

AI/ML: Deep learning, computer vision, self-supervised learning, semi-supervised learning, image segmentation, object detection, multimodal data integration, anomaly detection, time-series modeling

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, CAREamics, Captum

MLOps / Engineering: Git, GitLab CI/CD, DVC, Docker/containers, unit testing, TensorBoard monitoring, experiment tracking, model/version control, Conda, uv, Pixi

Scalable Computing: HPC-based ML workflows, Linux/Unix shell scripting, reproducible workflows

Robotics / Physical AI: Unity3D, Unreal Engine, motion capture, haptic devices, Leap Motion, sensor fusion, hand-motion replication, autonomous transport perception

Languages: Python, C#, C++, Java, SQL

SOFT SKILLS

Problem-solving, communication, cross-functional collaboration, independent working style, adaptability, agile mindset, and quality-focused approach; proficient in English (C1) and German (B2.1).

PROFESSIONAL EXPERIENCE

Helmholtz Zentrum Dresden Rossendorf (HZDR)

April 2025 - Present

Wissenschaftliche Mitarbeiterin (Machine Learning Engineer)

- Built and validated reproducible ML workflows for noisy, high-dimensional scientific imaging and tomography data, including self-supervised ROFEX sinogram denoising with CAREamics and deep learning-based tomography reconstruction, applying preprocessing, quality checks, experiment tracking, and documentation practices relevant to reliable AI model development in data-driven digital manufacturing environments.
- Developed and ran reproducible AI experimentation workflows on GPU/HPC clusters, using Git/GitLab, CI pipelines, DVC, containers, test cases, and structured validation workflows for scientific AI pipelines.
- Support interdisciplinary research teams through AI consulting, training through ERUM Hub, HPC Gateway Fall School, and HIDA, and practical guidance on machine learning workflows.
- Contributed to proposal writing, including an accepted **Helmholtz AI** funding application.
- Preparing an anomaly detection project for scientific datasets in collaboration with ROTOP Pharmaka GmbH, focusing on reliable feature development, validation, and model evaluation.
- Contributed to scientific writing, including [paper](#) co-authorship, an accepted **Helmholtz AI AIRE** funding proposal, and an accepted **HAICON 2026 poster** abstract.
- Presented ongoing tomography denoising research at **Building Bridges** and **Brains on Silicon** scientific conferences.
- Started evaluating an open-source Slurm queue-time prediction model for possible use on HZDR GPU/HPC clusters, focusing on training-data requirements, Slurm job-history aggregation, and reproducible benchmarking with RoSI or Hemera data.

Otto-von-Guericke-University Magdeburg (OVGU)

Feb 2025 - March 2025

Wissenschaftliche Mitarbeiterin (Machine Learning Engineer)

- Contributed to an AI-based interactive spine application using patient sensor data

Technische Universität Dresden (CeTI)

April 2021 - Jan 2025

Wissenschaftliche Hilfskraft mit abgeschlossener wissenschaftlicher Hochschulbildung

- Developed physical AI and robotics simulation workflows in Unreal Engine and Unity, leveraging C#, C++, VR/AR tools, Leap Motion, motion capture systems, and haptic devices for hand motion replication and collision-based feedback.
- Integrated and analyzed sensor data from motion capture (Qualisys, QTM Manager) and B&K sensors, measured latency to enhance precision, foster human-machine interaction, and advance the Internet of Skills.
- Upgraded the [FingerTac](#) wearable device from single to three-finger, integrating it into Unreal Engine in collaboration with [Dr. Thomas Hulin](#) from DLR, to improve augmented haptic feedback.
- Collaborated with cross-functional teams in an agile environment to integrate quality benchmarks, optimize data pipelines, validate system performance, and refine project goals for seamless system integration.

Technische Universität Dresden (ABSOLUT project)

January 2020 - June 2020

Wissenschaftliche Hilfskraft mit abgeschlossener wissenschaftlicher Hochschulbildung

- Researched ML-based object detection for safer autonomous transport systems.
- Developed stochastic occupancy grids and RANSAC-based detection for improved vehicle safety.
- Cleaned, transformed, and integrated stereo image data for accurate model training.
- Focused on data preprocessing, data transformation, and model validation for accurate object detection.

Lecturer (Bangladesh)

- Delivered lectures on C, Python, operating systems, databases, and machine learning.
- Supervised student thesis projects, providing mentorship and guidance.
- Promoted interdisciplinary collaboration through cross-departmental projects and participated in professional conferences to stay updated on cutting-edge research and trends in machine learning.
- Published peer-reviewed journal articles; full publication list available via [portfolio](#).

EDUCATION

Master of Science in Computational Modeling and Simulation (<i>Technische Universität Dresden</i>)	2019-2024
Focus (Visual Computing): Machine Learning, Computer Vision, Data Science, High-Performance Computing	
Master of Science in Computer Science and Engineering (<i>Jahangirnagar University, Bangladesh</i>)	2012-2013
Bachelor of Science with Honors in Computer Science and Engineering (<i>Jahangirnagar University, Bangladesh</i>)	2009-2012

PROJECTS**Self-supervised Denoising Of Raw Tomography Detector Data For Improved Image Reconstruction** 2025/26

- Developed CAREamics-based Noise2Void/N2V2 denoising workflows for noisy ROFEX sinogram/raw detector data before CT reconstruction.
- Prepared phantom, structured-packing, empty, dark, and hold-out scan datasets for masked-pixel training and structured validation.
- Evaluated reconstruction quality using PSNR, SSIM, CTF, MTF, residual/error maps, line profiles, and cross-validation across model depths and hyperparameters.
- Performed downstream proxy analysis for material peak separation and segmentation quality, showing cleaner reconstructions while preserving fine structure.
- Used Git, GitLab CI, test cases, and reproducible documentation to support reliable experimentation and model comparison.

AI-based interactive spine application using patient sensor data 2025

- Supported an AI-based Spine Interactive app for spinal disorders, using smartphone accelerometer and gyroscope data for mobility, gait-parameter analysis, and cognitive test integration.

Predict Baltic Sea Temperature and Salinity using two years of 6-hourly sampled data. 2024

- Enhance predictions using the Baltic Sea's real-life temporal and spatial data, leveraging time series data processing to surpass naive baselines (e.g., persistence, linear regression) with an ML model (LSTM + CNN).
- **Technology Used:** Python, PyCharm, scikit-learn, TensorFlow, CI/CD, DVC, Unit Testing, Container (Docker)

Deep-learning-based quantification of pancreatic texture using IR-Spectroscopy data 2023/24

- Developed a deep learning system using IR spectroscopy, real-life data collected from 30 patients at a university hospital, to analyze pancreatic tissue textures and tumor grades, aiding surgeons in assessing pancreatic cancer and post-operative fistula risk, under [Prof. Dr. Stefanie Speidel](#) at NCT, Dresden.
- Cleaned and integrated data from multiple sources to build robust classification datasets, using scratch-built and pre-trained ResNet18 models and MobileNetV1, achieving over 90% accuracy with ResNet18.
- Enhanced performance by integrating multimodal data (IR spectroscopic images and tumor grades) and applying dimensionality reduction techniques such as PCA to address dataset limitations, quality issues, and class imbalance.
- Applied findings from leading journals to develop methodologies, optimize algorithms for improved model performance, accuracy, and interpretability, and collaborated with oncologists on clinical applications.
- Implemented end-to-end ML workflows including data preprocessing, model training, experiment tracking, model versioning with DVC, CI/CD automation, monitoring with TensorBoard, and reproducibility practices using GitLab and PyTorch.
- **Technology used:** Python, NumPy, Pandas, Matplotlib, PyTorch, Captum, PCA, TensorFlow, Keras, PyCharm, ScikitLearn, CI/CD pipelines, and Git, applying both supervised and semi-supervised learning techniques.

Understanding Deep-Learning requires rethinking generalization 2020/21

- Explored deep neural network architectures' superior generalization compared to traditional regularization techniques, conducting experiments with AlexNet, VGG19, and ResNet.
- Achieved near-zero training errors with noisy data, with performance drops on shuffled training/testing data and random labels.
- Demonstrated the impact of regularization (early stopping, batch normalization) and optimization techniques (quantization, pruning) on generalization, emphasizing the need for new strategies to manage generalization errors.
- **Technology used:** Python, TensorFlow, Keras, JupyterLab, ScikitLearn.

Exploring Medical Data with AR and Spatially Tracked Tablet 2020

- Developed a Unity3D simulation in C# for HoloLens and Tablet interaction with 3D liver CT images.
- Enabled volumetric data creation, spatial navigation, and gesture-based interaction for enhanced organ visualization.
- Developed HoloLens-based gesture interaction with a liver model, enabling selection of specific CT regions for operation planning.

Kidney Segmentation in Abdominal CT Images Using Fuzzy Logic and Region Growing 2017

- Developed a fuzzy logic-based filtering technique to enhance abdominal CT images and preserve edges.
- Applied region-growing algorithm for kidney segmentation, refining with ROI identification, pixel filling, erosion, labeling, and dilation.
- Achieved 73% accuracy and a PSNR of 26.09 dB, outperforming traditional methods in quality and speed.

Object-Oriented Programming with Java 2022

- Gained object-oriented programming skills through the SCPROG course at TU Dresden, focusing on Java.
- Completed projects on core Java OOP concepts, including inheritance, polymorphism, interfaces, and generics.
- Collaborated with peers to enhance problem-solving and debugging skills.
- Utilized GitLab for version control, improving project management and teamwork.